A REPRESENTABLE MORPHISM OF AFFINE FORMAL ALGEBRAIC SPACES IS REPRESENTABLE BY SCHEMES AND AFFINE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file formal-spaces.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- Affineness of a representable transformation can be checked locally on the target. Morphisms of Spaces, Lemma spaces-morphisms-lemma-affine-local, invoked at proof lines 4215-4217. Reduces representability by schemes and affineness to affineness of the base changed morphisms.
- Affine formal algebraic spaces as filtered colimits of affine schemes, and factorisation of morphisms through a stage. Same chapter, Definition definition-affine-formal-algebraic-space at proof lines 4218-4219 and Lemma lemma-factor-through-thickening at proof lines 4227-4229. Supply the presentations and the first factorisation.
- A representable morphism of affine formal algebraic spaces is quasi-compact. Same chapter, Lemma lemma-characterize-quasi-compact-morphism, invoked at proof lines 4229-4231. Makes the fibre product quasi-compact so that the second factorisation is available.
- The fibre product of affine schemes is affine. Standard, used at proof lines 4234-4236. Gives affineness of the fibre product after the two factorisations.

Source reference. Original definition 03XZ: sources/stacks-chow/tags.txt, source lines 5166–5166. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 0AJB: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 3803–3803. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 0AIA: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 2080–2080. Pointer outside the counted span, invoked once

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for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 03WG: sources/stacks-chow/tags.txt, source lines 5111–5111. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original remark 0AIS: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 2769–2769. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Definition 0.1. Let S be a scheme. We say a sheaf X on $(Sch/S)_{fppf}$ is an *affine formal algebraic space* if there exist

(1) a directed set Λ,
(2) a system $(X_\lambda, f_{\lambda\mu})$ over Λ in $(Sch/S)_{fppf}$ where
    (a) each X_λ is affine,
    (b) each $f_{\lambda\mu} : X_\lambda \to X_\mu$ is a thickening,

such that

$$X \cong \operatorname{colim}_{\lambda\in\Lambda} X_\lambda$$

as fppf sheaves and X satisfies a set theoretic condition (see Remark 0AIS). A *morphism of affine formal algebraic spaces* over S is a map of sheaves.

Lemma 0.2. *Let S be a scheme. Let $f : X \to Y$ be a morphism of affine formal algebraic spaces which is representable by algebraic spaces. Then f is representable (by schemes) and affine.*

Proof. We will show that f is affine; it will then follow that f is representable and affine by Morphisms of Spaces, Lemma 03WG. Write $Y = \operatorname{colim} Y_\mu$ and $X = \operatorname{colim} X_\lambda$ as in Definition 0.1. Let $T \to Y$ be a morphism where T is a scheme over S. We have to show that $X \times_Y T \to T$ is affine, see Bootstrap, Definition 03XZ. To do this we may assume that T is affine and we have to prove that $X \times_Y T$ is affine. In this case $T \to Y$ factors through $Y_\mu \to Y$ for some μ, see Lemma 0AIA. Since f is quasi-compact we see that $X \times_Y T$ is quasi-compact (Lemma 0AJB). Hence $X \times_Y T \to X$ factors through X_λ for some λ. Similarly $X_\lambda \to Y$ factors through Y_μ after increasing μ. Then $X \times_Y T = X_\lambda \times_{Y_\mu} T$. We conclude as fibre products of affine schemes are affine. □